

Introduction to Food Science and Nutrition

Science

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Introduction to Food Science and Nutrition (A07061)

Short Title:	Introduction to Food Science and Nutrition	
Faculty	Science and Computing	
Department	Science	
Cluster	Food Science	
Author	EDUGGAN	
Credits	5	Level: Introductory

Description of Module / Aims

This module will provide the student with an introduction to key aspects of food science: food chemistry, food safety, food quality and processing and nutrition. Issues of importance affecting the food industry will be reviewed, including consumer trends and health.

Programmes

SCIE-0039	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / E
SCIE-0039	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 2 / E
SCIE-0039	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / E
SCIE-0039	BSc (Hons) in Physics for Modern Technology (WD_KPHE_B)	1 / 2 / E
SCIE-0039	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 2 / E
SCIE-0039	BSc in Agriculture (WD_SAGUL_D)	2 / 4 / E
SCIE-0039	BSc in Food Science with Business (WD_SFDSC_D)	1 / 2 / M

Indicative Content

- Food Science: a multidisciplinary subject and role of nutrition in supporting human health
- Structure of the Irish food industry and its contribution to the economy: employment, production, exports, domestic consumption
- Agencies involved in food and health policy formation, food promotion and regulation
- Current issues presenting threats and opportunities to the food industry
- Basic techniques used in a food science laboratory

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Recognise the scope of Food Science and the linkages with other related disciplines.
2. Illustrate the importance of the food industry to the Irish economy and the various food regulation and promotion agencies.
3. Identify current issues presenting threats and opportunities to the food industry.
4. Employ a variety of laboratory techniques relevant to the broad discipline of food science.
5. Report on a range of laboratory techniques for a variety of food types.

Learning and Teaching Methods

- Lectures on indicative content will be delivered.
- Guest lectures and videos will be used to support the lectures, and to focus on case studies of relevance.

- Class discussions on topical issues.
- Laboratory practicals will introduce the students to basic experimental techniques used in food science.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	40%	1,2,3
<i>Practical</i>	60%	4,5

Assessment Criteria

- <40%:** Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.
- 40%-49%:** Will be able to show limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.
- 50%-59%:** As well as the above, the student will be to show reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.
- 60%-69%:** In addition, the student will demonstrate more detailed knowledge of the all topics contained in the learning outcomes and display good competence with regards to use of associated laboratory instrumentation and very good laboratory practice.
- 70%-100%:** All of the above to excellent levels. In addition the learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	24	
Independent Learning	99	

Supplementary Material

- "Department of Agriculture, Food and the Marine." <http://www.agriculture.gov.ie/>
- "European Food Safety Authority." <http://www.efsa.europa.eu/>
- "Food Safety Authority of Ireland." <https://www.fsai.ie/>

Requested Resources

- Room Type: Chemistry Lab